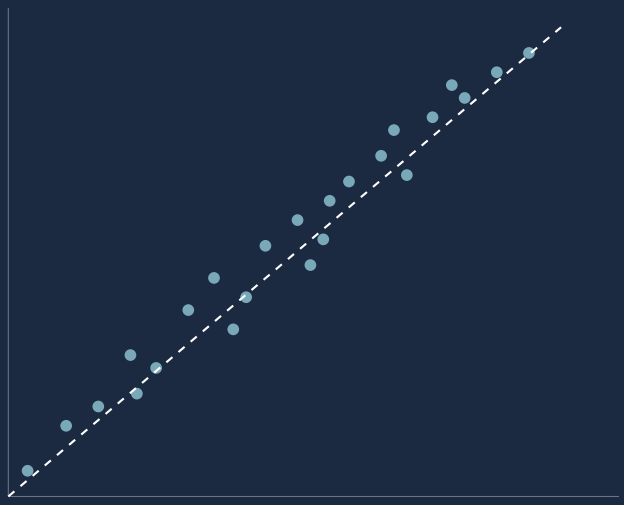
A COMPUTATIONAL MATERIALS STUDY



Physics-Informed Machine Learning of Battery Electrode Voltages

From Composition to Cross-Ion Transfer

Shaista Saddique

Department of Physics
Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia

October 2026

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From Composition to Cross-Ion Transfer

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Data: Materials Project insertion-electrode collection (database version 2026.09.28), licensed CC BY 4.0. Software: Python, pymatgen, mp-api, XGBoost, scikit-learn, CHGNet, Gradio. All computations were carried out on Google Colab.

Typeset in TeX Gyre Pagella and TeX Gyre Heros with LaTeX.

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Abstract

Computing the voltage of a single battery electrode with density functional theory (DFT) takes hours to days of cluster time. This work shows that a machine-learning model trained on existing calculations can predict electrode voltages in a fraction of a second, and that a single physical correction allows it to make useful predictions for working ions it has never seen.

A gradient-boosting model was trained on 2,634 DFT-computed insertion electrodes from the Materials Project, covering six working ions: Li, Na, K, Mg, Ca and Zn. Each stage of the project added one piece of physics and was tested on framework-based splits, so that no host material is shared between training and test sets. The final model predicts the average voltage of unseen host materials with a mean absolute error (MAE) of 0.368 ± 0.019 V ($R^2 = 0.82$).

Table 1: *Progression of model accuracy. Test MAE is the mean ± standard deviation over 10 framework-grouped splits.*

Stage	What was added	Test MAE (V)
Reference	Mean voltage of each ion	0.884 ± 0.021
Chapter 2	Composition features of host and ion	0.445 ± 0.022
Chapter 3	Oxidation states from charge balance	0.391 ± 0.016
Chapter 4	CHGNet crystal-structure fingerprints	0.368 ± 0.019

Transfer between ions is the central result. Monovalent Na and K transfer without loss of accuracy, while Li, Ca and Zn fail with errors near 1 V. The failure is almost entirely a constant offset set by each ion's metal reference potential. Training on the host potential $V + E^\circ$, where E° is the ion's standard electrode potential, removes 96–108% of that offset error without any data from the target ion and halves the mean zero-shot error from 0.81 to 0.44 V, at no cost to in-distribution accuracy. The final model is packaged as an interactive web application, the *Electrode Voltage Predictor*.

Abbreviations and Symbols

CHGNet	Crystal Hamiltonian Graph Neural Network, a pretrained universal interatomic potential
DFT	Density functional theory
E°	Standard electrode potential of a working ion (V vs SHE)
E_{host}	Host potential, $V + E^\circ$, the physics-informed training target
GGA, GGA+U	Generalised gradient approximation (with Hubbard correction)
GNN	Graph neural network
MAE	Mean absolute error
MP	Materials Project
R^2	Coefficient of determination
SHE	Standard hydrogen electrode
V	Average electrode voltage vs the working-ion metal
XGBoost	Extreme gradient boosting, a tree-ensemble regression method
Zero-shot	Prediction for a working ion absent from training
Few-shot	Prediction after adding a small share of the target ion's data to training

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Building the Electrode Dataset

KEY RESULT

The Materials Project provides 6,847 DFT-computed insertion electrodes. After cleaning, 2,634 electrodes for six working ions form the dataset used throughout this work.

1.1 Motivation

Computing one electrode voltage with DFT takes hours to days of cluster time, because the energies of the host with and without the inserted ions must both be calculated. A model that learns from existing calculations can screen thousands of candidates in seconds. Most existing data are for lithium, while sodium, potassium and magnesium batteries use cheaper, more abundant elements and have far less data. This gap motivates the transfer question studied in Chapter 5.

1.2 Data source and download

Each entry in the Materials Project insertion-electrode collection is a host framework (for example $FePO_4$) with a working ion inserted step by step. For each electrode, the average voltage, gravimetric and volumetric capacity, energy density, volume change, thermodynamic stability of the charged and discharged states, and the host crystal structure were downloaded through the `mp-api` client (database version 2026.09.28). The client returns an Arrow dataset; reading it directly with `pyarrow` loaded all 6,847 records in seconds, whereas row-by-row conversion took more than 15 minutes.

1.3 Cleaning

Four rules, summarised in Table 1.1, were applied to obtain a reliable training set.

For modelling, six focus ions with enough data were kept: Li (1,351 electrodes), Mg (384), Ca (321), Na (287), Zn (173) and K (118), a total of 2,634. Y, Rb, Al and Cs (about 55–70 electrodes each) were set aside.

1.4 Exploration of the data

Lithium accounts for about half of the data (Figure 1.1). Li, Na, K and Ca have median voltages of about 3.4–3.8 V, while Mg ($\approx$ 2.6 V), Al ($\approx$ 2.0 V) and Zn ($\approx$ 1.6 V) are clearly lower,

Table 1.1: *Data-cleaning rules applied to the raw Materials Project electrodes.*

Rule	Reason
Drop entries missing voltage, capacity or structure	Every model input must be complete
Keep $0\,\mathrm{V} \leq$ average voltage $\leq 6\,\mathrm{V}$	Removes unphysical values from unstable phases (e.g. a $LiFePO_4F$ entry at 6.84 V)
Keep discharged states with energy above hull $\leq 0.1\,\mathrm{eV/atom}$	Trains on realistic, near-stable materials
Remove duplicate framework + ion pairs	Keeps the most stable entry for each pair

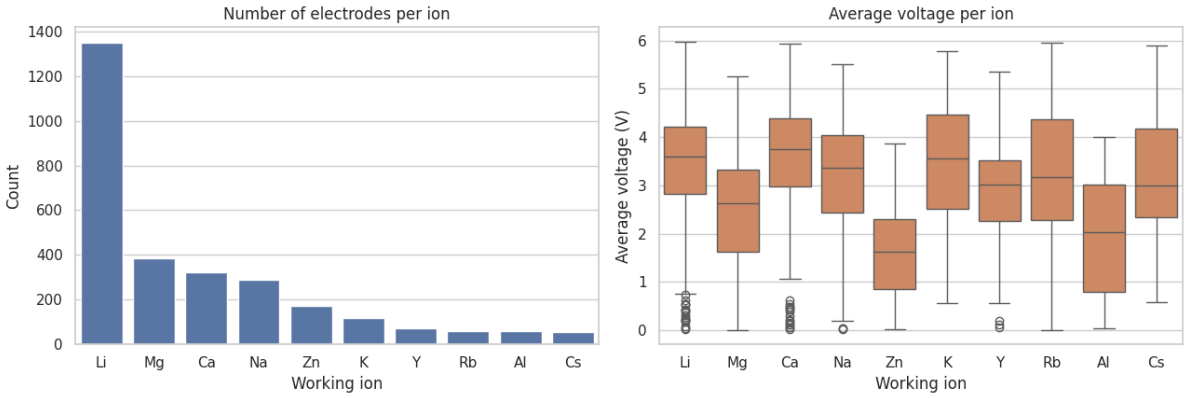


Figure 1.1: *Left: number of electrodes per working ion after cleaning. Right: distribution of the average voltage for each ion.*

consistent with their less negative standard electrode potentials. Points near 0 V correspond to anode-like hosts.

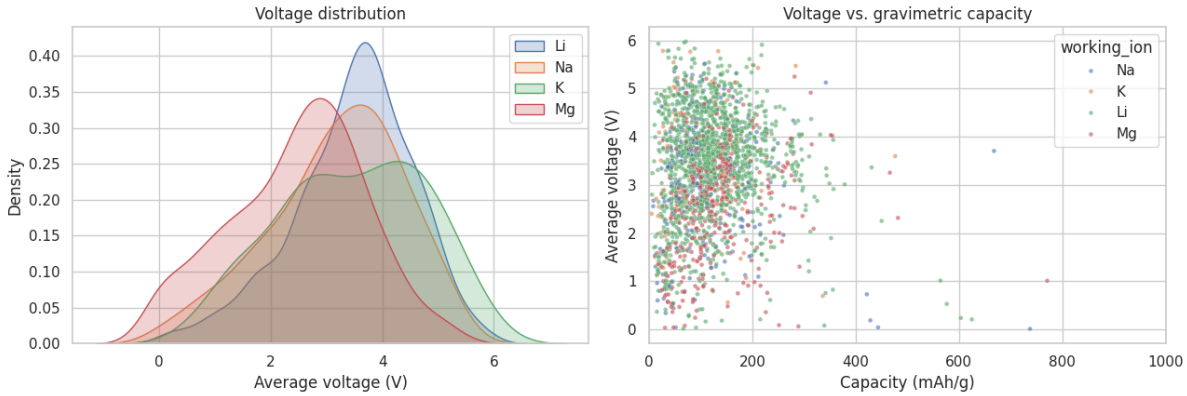


Figure 1.2: *Left: voltage distributions for Li, Na, K and Mg. Right: average voltage against gravimetric capacity.*

Li peaks sharply near 3.7 V; Na has a similar but slightly lower distribution; K is much broader, with a tail to 5–6 V; and Mg is shifted to about 2.8 V (Figure 1.2). Most electrodes have capacities below 300 mAh g^{-1}, and voltage and capacity are only weakly correlated. The few hosts above 400 mAh g^{-1} sit at low voltage, the familiar capacity–voltage trade-off.

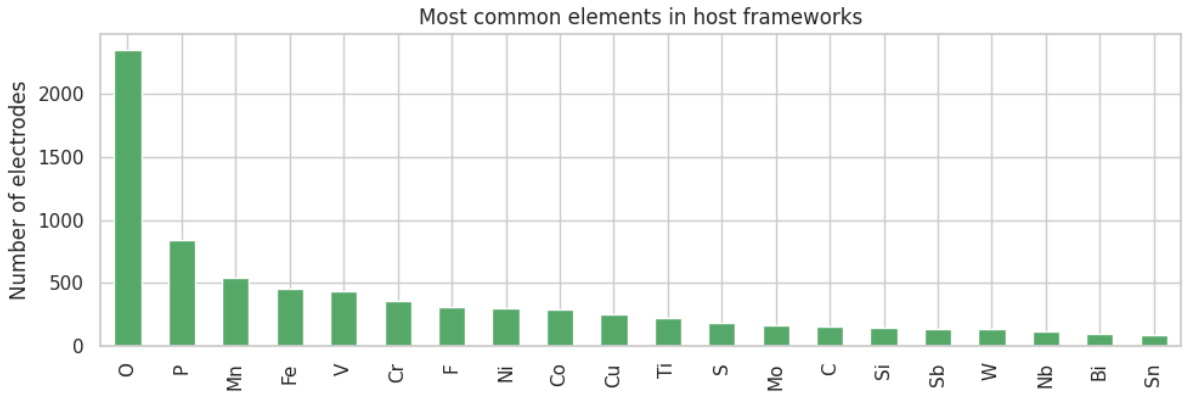


Figure 1.3: *The 20 most common elements in the host frameworks.*

Oxygen appears in about 2,350 electrodes, so oxide cathodes dominate (Figure 1.3). Phosphorus ($\approx$ 840) marks polyanion cathodes, often combined with fluorine ($\approx$ 300). Mn, Fe, V, Cr, Ni, Co, Cu and Ti are the redox-active metals, while C, Si, Sb, Sn and Bi mark anode and alloying hosts.

1.5 Implications for modelling

- Ion-specific features are needed, because each ion's voltage distribution is shifted differently.
- The same framework often appears with several ions, so train/test splits must keep each framework entirely on one side.
- The model will be most reliable for oxide and phosphate cathodes and weaker for sulfides, anodes and rare chemistries.

A Composition-Only Baseline

KEY RESULT

From the chemical formula alone, an XGBoost model predicts the voltage of unseen host frameworks with an MAE of 0.45 V, half the error of the per-ion mean (0.88 V).

2.1 Features

Every electrode is described by numbers computed only from the host formula and the working ion, so the model can be applied to materials that have never been calculated (Table 2.1).

Table 2.1: *Feature groups used by the composition-only model.*

Group	Features
Host composition statistics	Atom-fraction-weighted mean, standard deviation, minimum and maximum of electronegativity X, atomic radius, atomic number Z, atomic mass, periodic row and group, Mendeleev number and first ionisation energy
Host chemistry	Transition-metal fraction and mean Z of the transition metals; fractions of O, F, S and other anions; polyanion flag (P, S, Si, B, As with O)
Working ion	Charge, ionic radius, electronegativity, ionisation energy, mass
Insertion amount	Fraction of the ion in the discharged formula (`frac_ion_discharge`)

2.2 Splitting without leakage

The same framework (e.g. $FePO_4$) often appears with several ions. A random split would let the model see $FePO_4$ with Na during training and then test it on $FePO_4$ with Li, which overstates accuracy. All splits therefore use `GroupShuffleSplit` and `GroupKFold` grouped by `framework_formula`, so every framework lies entirely in the training set or entirely in the test set (80/20).

2.3 Model and reference

XGBoost (gradient-boosted decision trees) was trained on the 80% training frameworks, with hyperparameters chosen by grouped cross-validation. As a reference, predicting each ion's

mean voltage gives an MAE of 0.884 V.

2.4 Results

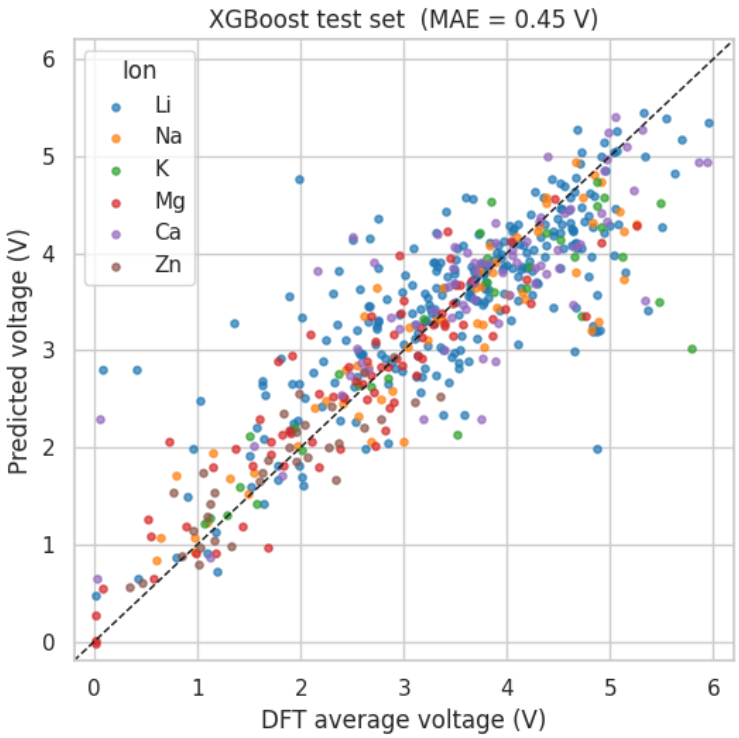


Figure 2.1: *Predicted against DFT voltage for the test frameworks, coloured by working ion* ($MAE = 0.45\,V$).

Points follow the diagonal from 0 to 6 V for all six ions (Figure 2.1). Over 10 repeated splits the MAE is 0.445 ± 0.022 V. The largest errors are a few anode-like hosts predicted near 3 V when DFT gives about 0 V (or the reverse), and high-voltage Li and K hosts above 5 V, which are predicted too low: the model compresses the extremes towards the middle of the distribution.

2.5 What the model learned

The maximum electronegativity in the host (`host_X_max`, 36%) dominates (Figure 2.2). It separates oxides and fluorides (high X, high voltage) from sulfides, carbides and alloys (low X, low voltage), the inductive effect familiar from polyanion cathodes. The minimum atomic radius (17%), maximum Mendeleev number and maximum ionisation energy encode the same anion chemistry. Ion radius and ion electronegativity carry the ion dependence, and `host_frac_F` and `host_polyanion` capture the higher voltage of fluorinated and polyanion frameworks.

2.6 Limitation identified

The model knows which elements are present but not how many electrons the redox metal gives up. $FePO_4$ ($Fe^{3+} \rightarrow Fe^{2+}$) and a host containing Fe^{4+} look similar to it, although their voltages differ. This motivated adding oxidation states in Chapter 3.

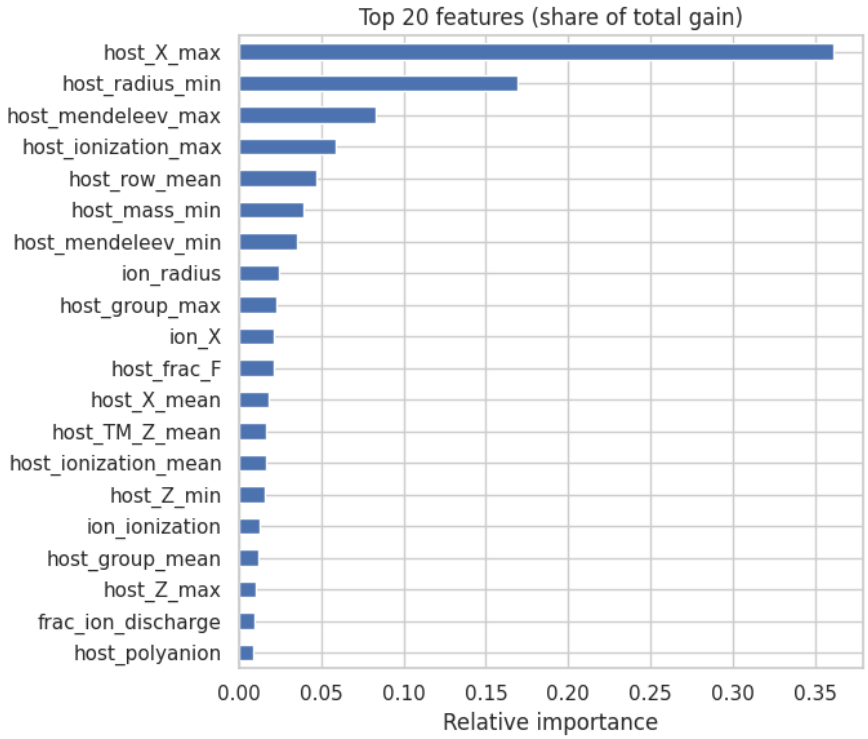


Figure 2.2: *The 20 most important features, by share of total gain.*

CHAPTER 3 WEEK 2B

Teaching the Model Redox Chemistry with Oxidation States

> **KEY RESULT**
>
> **Adding oxidation states from charge balance lowered the MAE from 0.445 to 0.391 V, a 12% improvement and the largest single gain after the baseline.**

3.1 Idea

The voltage comes from the redox couple of the transition metal, for example Fe^{3+}/Fe^{2+} near 3.4 V vs Li in $LiFePO_4$, or Co^{4+}/Co^{3+} near 4 V in $LiCoO_2$. If the average oxidation state of the redox metal in the charged and discharged host is known, the model can learn which couple is active.

3.2 Charge-balance rules

The average oxidation state of the redox metals is solved from charge neutrality, using fixed charges for all other elements (Table 3.1).

Table 3.1: *Fixed charges used to solve for the redox-metal oxidation state.*

Element type	Charge used
Anions	O −2; F −1; S, Se, Te −2; N −3; Cl, Br, I −1
Oxo-anion centres (when O is present)	P +5, Si +4, B +3, As +5, C +4, S +6, Se +6, N +5, Cl +7, I +5, Te +6, Ge +4
Fixed cations	H +1; alkali and alkaline-earth metals; Zn and Cd +2; Al, Ga, Sc, Y and lanthanides +3
Post-transition redox elements	Sb, Sn, Bi, Pb, Tl, In treated as redox-active

The new features are the oxidation state in the charged and discharged host (`ox_charged`, `ox_discharged`), their change (`ox_change`), the metal's common maximum oxidation state (`ox_common_max`), how far the charged state exceeds it (`ox_excess`, flagged as `ox_above_max` when greater than 0.2), and flags for unreliable balances, hosts without a redox metal, elemental hosts and anion-free hosts.

3.3 Debugging unphysical values

The first version produced impossible oxidation states such as -298 and $+31$, and flagged 977 electrodes. Three causes were found and fixed:

1. Hydrogen was not counted; it is now fixed at $+1$.
2. Oxo-anion centres such as P in PO_4 were treated as redox metals; they are now fixed at their group valence.
3. Results outside -4 to $+8$, or changes outside 0–4 electrons per metal, are now set to missing and marked `ox_unreliable`.

After the fix, 924 electrodes are flagged above the common maximum (mostly highly oxidised Li-rich and Co/Ni oxides, a genuine chemical signal) and only 43 are unreliable.

3.4 Results

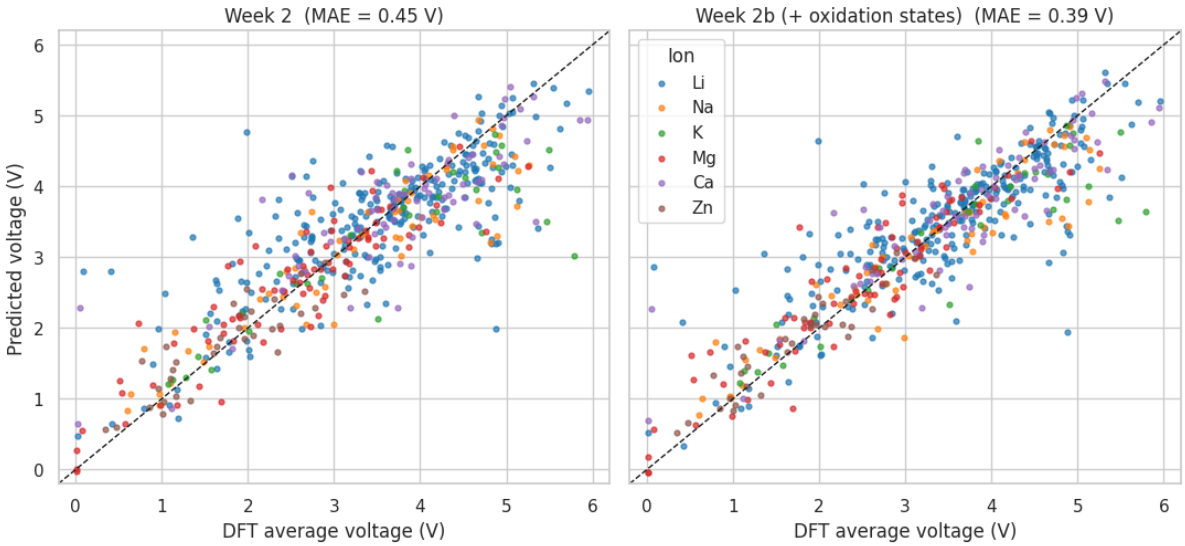


Figure 3.1: *Parity plots before (left, MAE 0.45 V) and after (right, MAE 0.39 V) adding the oxidation-state features.*

The cloud tightens around the diagonal, most visibly between 2 and 4 V where the transition-metal cathodes sit (Figure 3.1). Over 10 splits the MAE falls from 0.445 ± 0.022 to 0.391 ± 0.016 V, and the spread between splits also shrinks. Anode-like outliers near 0 V and a few K and Li hosts above 5 V remain.

`host_X_max` is still first, but its share drops from 36% to 27% and the importance spreads more evenly over the anion descriptors (Figure 3.2). `ox_above_max` enters the top twelve: hosts with the redox metal above its usual maximum (Co^{4+}, Ni^{4+}, or anion redox) give high voltages, which composition statistics alone could not express. `ox_common_max` also appears.

LESSON

A physically motivated descriptor helped only after its implementation had been checked against chemical sense. Inspecting extreme values before training was essential.

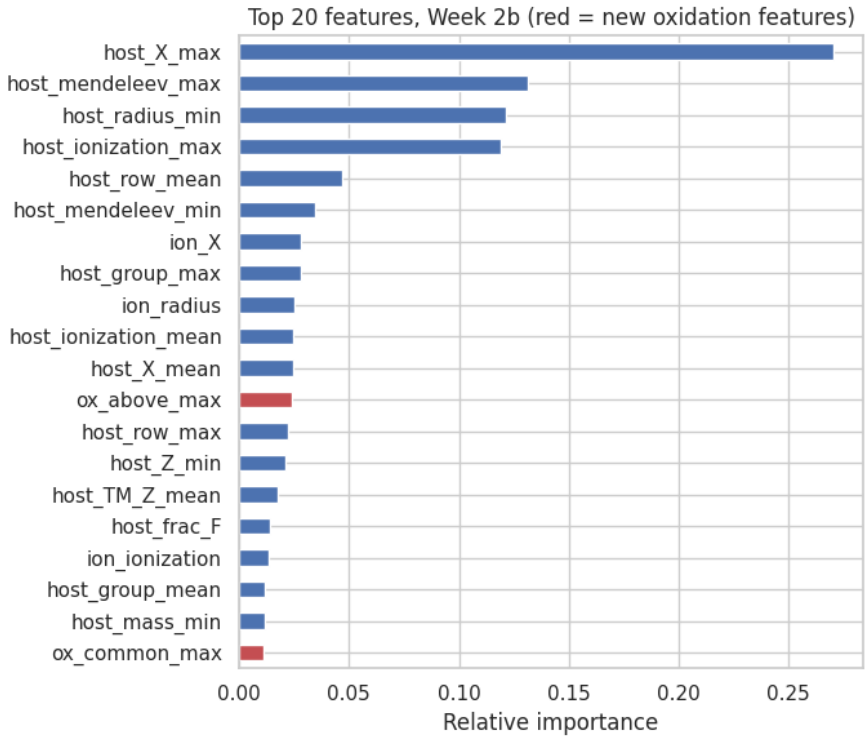


Figure 3.2: *The 20 most important features after adding oxidation states; new features are shown in red.*

CHAPTER 4 WEEK 3

Adding Crystal Structure with a Pretrained Universal Potential

> **KEY RESULT**
>
> **Combining composition and oxidation features with 64-dimensional CHGNet crystal fingerprints gives the best model of this work: MAE 0.368 ± 0.019 V ($R^2 = 0.82$).**

4.1 Why structure matters

Polymorphs with the same formula (layered, spinel or olivine) can differ in voltage, and the ion's site and the openness of the framework matter. Composition features cannot see these differences.

4.2 Method

CHGNet is a graph neural-network interatomic potential pretrained on about 1.5 million DFT calculations from the Materials Project. For each charged host structure, its last-layer crystal representation was extracted with `predict_structure(…, return_crystal_feas=True)["crystal_fea"]`, a 64-number fingerprint of the local bonding environment. No CHGNet training was needed, so this step ran on the free Colab GPU in minutes. Three models were compared on the same framework-grouped splits:

1. Chapter 3 features (composition and oxidation states);
2. CHGNet fingerprint and working-ion features;
3. Combined: both feature sets together.

4.3 Results

The structure fingerprint alone (MAE 0.40 V) performs as well as the hand-built chemistry features (0.39 V), even though CHGNet was never trained on voltages (Figure 4.1). A pretrained universal potential therefore already encodes much of the redox chemistry. The combined model is best (0.37 V): the cloud is tighter between 2 and 5 V, and fewer high-voltage hosts are underpredicted. The two feature sets are complementary, because charge balance captures the electron count while CHGNet captures the local structure.

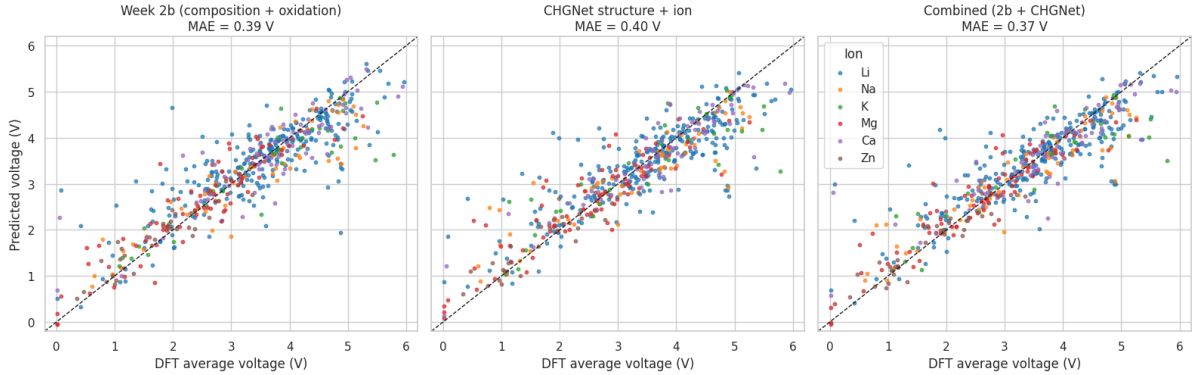


Figure 4.1: *Parity plots for the three models on the same test frameworks.*

4.4 Remaining errors

The largest errors are still at the extremes: anode-type hosts near 0 V (graphite and other carbon or alloy hosts, which have no transition metal and are rare in the data) and a few Li and K hosts above 5 V. These limits are flagged in the application described in Chapter 7.

Robustness and Transfer Between Working Ions

KEY RESULT

Over 10 framework splits every added piece of physics gives a reliable gain. A model asked to predict an ion it has never seen, however, fails for Li, Ca and Zn with errors near 1 V, almost entirely through a constant offset.

5.1 Repeated splits

A single 80/20 split can be lucky. Every model was therefore retrained on 10 different framework-grouped splits.

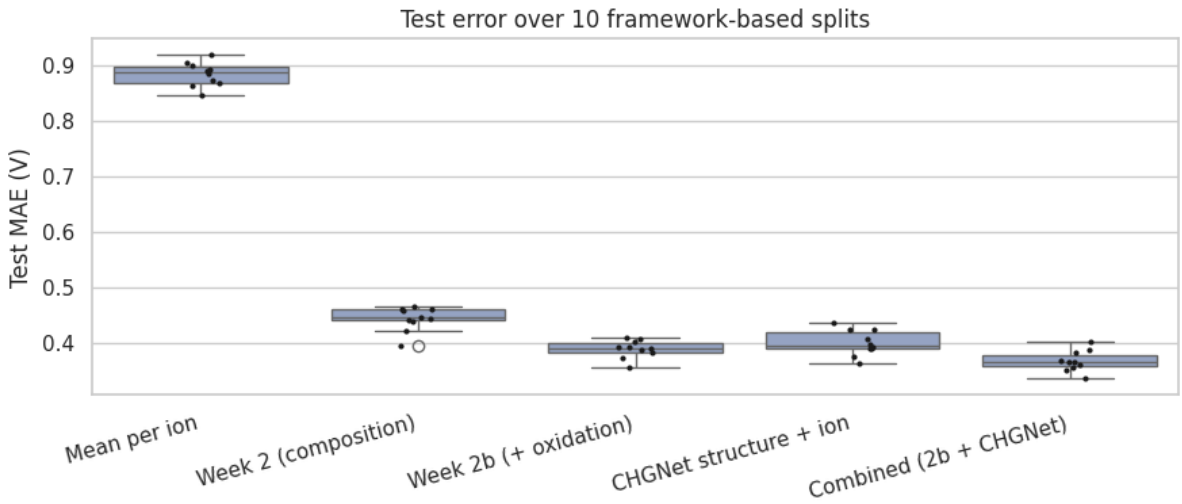


Figure 5.1: *Test MAE of each model over 10 splits; each dot is one split.*

Table 5.1: *Test MAE of each model over 10 framework-grouped splits.*

Model	MAE (V), mean ± SD
Mean voltage per ion	0.884 ± 0.021
Composition (Chapter 2)	0.445 ± 0.022
+ oxidation states (Chapter 3)	0.391 ± 0.016
CHGNet structure + ion	≈ 0.40
Combined (Chapter 4)	0.368 ± 0.019

The boxes in Figure 5.1 barely overlap, so each improvement is larger than the split-to-split noise.

5.2 Zero-shot transfer: an ion never seen in training

For each ion in turn, all its electrodes were removed from training and the model was tested on them. Two settings were used: *known hosts*, where the framework had been seen with other ions, and *new hosts* (*strict*), where the framework had never been seen at all. Settings with fewer than 100 training electrodes were skipped.

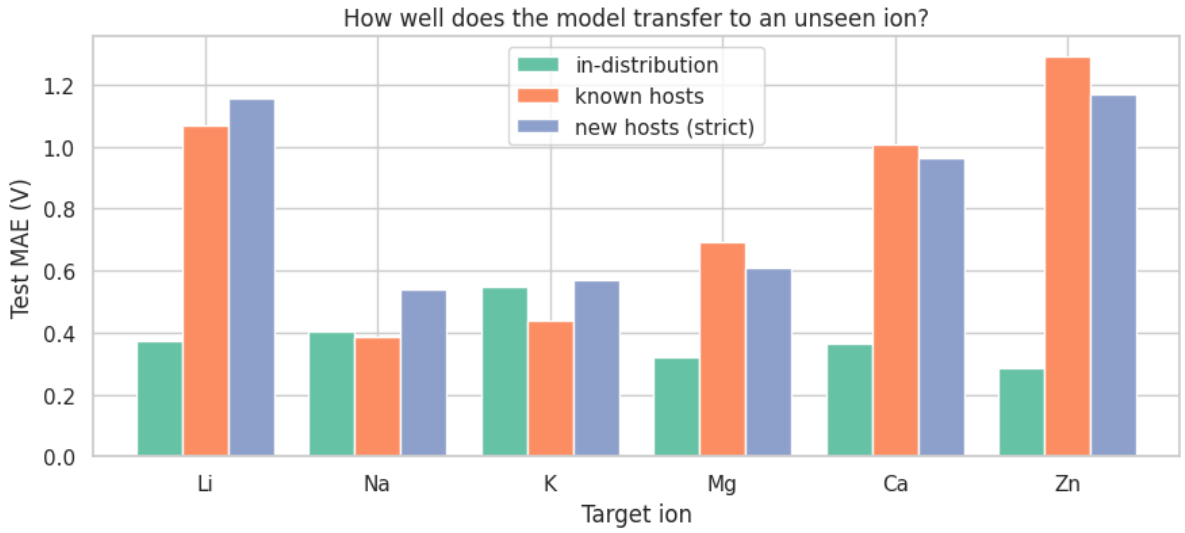


Figure 5.2: *MAE in-distribution, zero-shot on known hosts, and zero-shot on new hosts, for each target ion.*

Na and K transfer almost for free (Figure 5.2): Na 0.39 V and K 0.44 V on known hosts, close to or better than their in-distribution errors. Mg is moderately worse (0.69 V), while Li (1.07 V), Ca (1.01 V) and Zn (1.29 V) fail. New hosts are usually harder still, as expected.

5.3 The failure is an offset

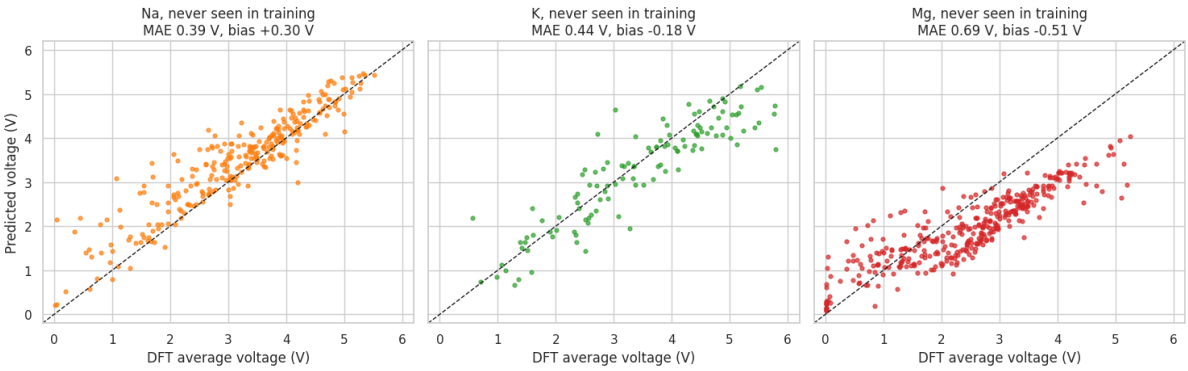


Figure 5.3: *Zero-shot parity for Na (bias +0.30 V), K (−0.18 V) and Mg (−0.51 V).*

The Mg points form a clean line parallel to the diagonal but shifted down (Figure 5.3). The model ranks the hosts correctly but places the whole ion at the wrong voltage level. The same

pattern appears for Li (bias −0.98 V), Ca (−0.93 V) and Zn (+1.27 V). The voltage of an ion is measured against its own metal, so each ion has its own zero point, and the model can only guess that zero point for an ion it has never seen.

5.4 Few-shot learning

A small share of the target ion's frameworks (0–50%) was added back to training, and the error was measured on the rest.

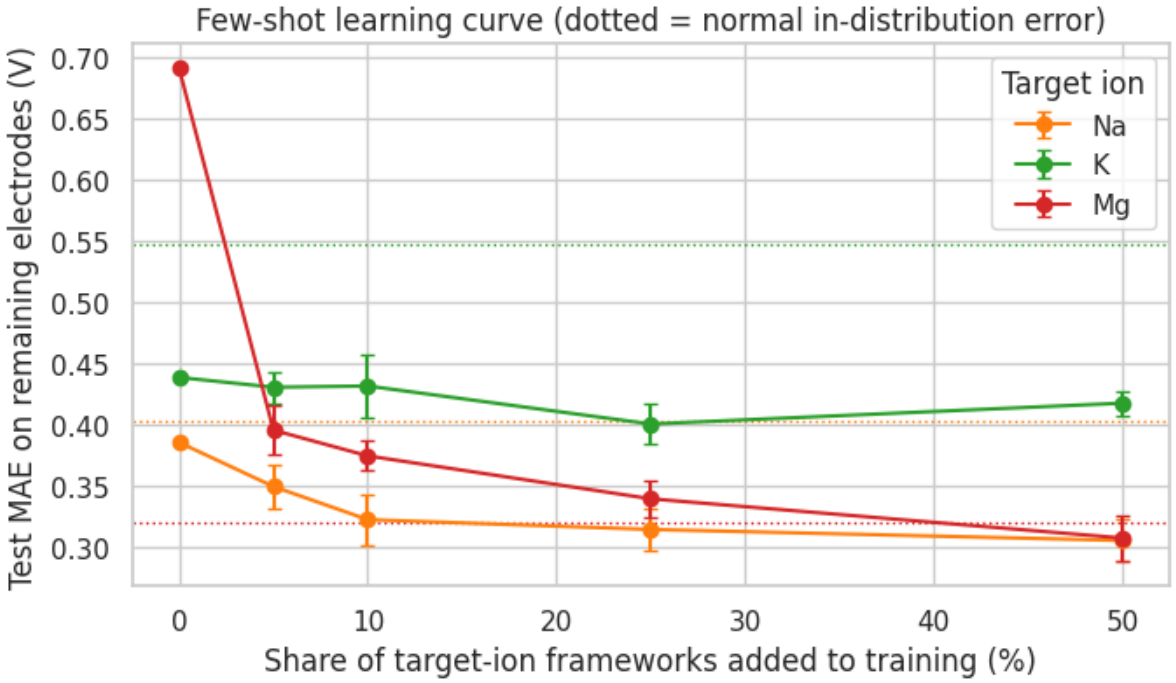


Figure 5.4: *Test MAE as target-ion frameworks are added to training; dotted lines show the in-distribution error.*

Adding only 5% of the Mg frameworks cuts its error from 0.69 to about 0.40 V, and 50% brings it to 0.31 V, at its in-distribution level (Figure 5.4). Na improves from 0.39 to 0.31 V. K stays near 0.42 V, already below its in-distribution error of 0.55 V, because it benefits from the larger Na and Li sets. A few calculations for a new ion are enough to fix the offset. This suggests the offset could be removed with no calculations at all if its physical origin were built into the model, which is the subject of Chapter 6.

CHAPTER 6 WEEK 4B

A Physics-Informed Reference Shift for Cross-Ion Transfer

KEY RESULT

Training on the host potential $E_{\text{host}} = V + E^{\circ}(\text{ion})$ instead of the voltage V halves the zero-shot error (mean over six ions: 0.81 $\rightarrow$ 0.44 V) and removes 96–108% of the offset error for Zn, Mg, Li and Ca, at no cost to in-distribution accuracy.

6.1 Physical basis

An electrode voltage V is measured against the metal of the working ion (Li/Li^{+}, Zn/Zn^{2+}, ...). Different ions therefore use different zero points. Shifting each voltage by the ion's standard electrode potential E° places all ions on one common scale, referenced to the standard hydrogen electrode:

$$E_{\text{host}} = V + E^{\circ}(\text{ion}), \qquad V_{\text{pred}} = E_{\text{host,pred}} - E^{\circ}(\text{ion})$$

Table 6.1: *Standard electrode potentials of the working ions.*

Ion	E° (V vs SHE)
Li	−3.04
K	−2.93
Ca	−2.87
Na	−2.71
Mg	−2.37
Zn	−0.76

The model then learns a property of the host, its redox potential, and needs only the tabulated E° of a new ion. This explains the offsets in Chapter 5: Zn, whose E° is about 2 V higher than those of the alkali metals, was overpredicted by 1.27 V, while Li, the most negative, was underpredicted by 0.98 V.

6.2 Upper bound: the oracle offset

To judge the correction fairly, an *oracle* was computed: the direct zero-shot predictions shifted by the best possible constant, found from the test data itself. This is the best any constant correction could achieve.

6.3 Results

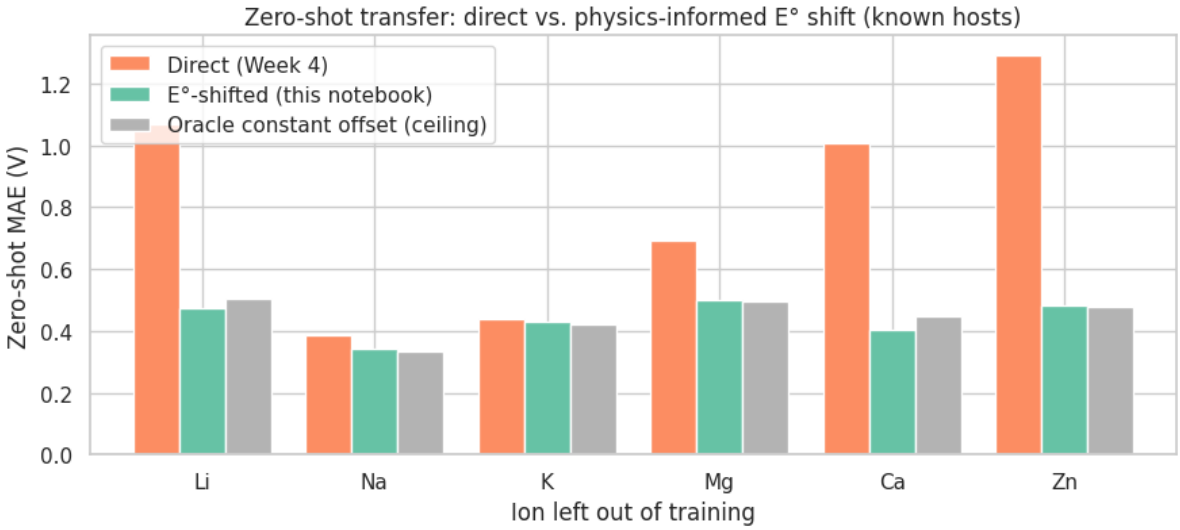


Figure 6.1: *Zero-shot MAE on known hosts: direct prediction (Chapter 5), E°-shifted prediction, and the oracle constant offset.*

Table 6.2: *Zero-shot MAE on known hosts for each ion left out of training.*

Ion left out	Direct (V)	E°-shifted (V)	Oracle (V)
Li	1.07	**0.47**	0.50
Na	0.39	**0.34**	0.33
K	0.44	**0.43**	0.42
Mg	0.69	**0.50**	0.49
Ca	1.01	**0.40**	0.45
Zn	1.29	**0.48**	0.48
Mean	0.81	**0.44**	0.45

The E° shift reaches the oracle ceiling for every ion (Table 6.2). For Li and Ca it is slightly better than the oracle, hence more than 100% of the offset error is removed, because the shift also changes what the trees learn rather than only moving the predictions. No data from the target ion are used.

The Zn bias falls from $+1.27$ to -0.40 V, Ca from -0.93 to -0.13 V, and Li from -0.98 to $+0.17$ V (Figure 6.2). The point clouds move onto the diagonal while keeping their shape, confirming that the error was an offset. The remaining Zn bias reflects that Zn^{2+} chemistry (a d^{10} ion in mostly aqueous-type hosts) differs from that of the alkali and alkaline-earth training ions.

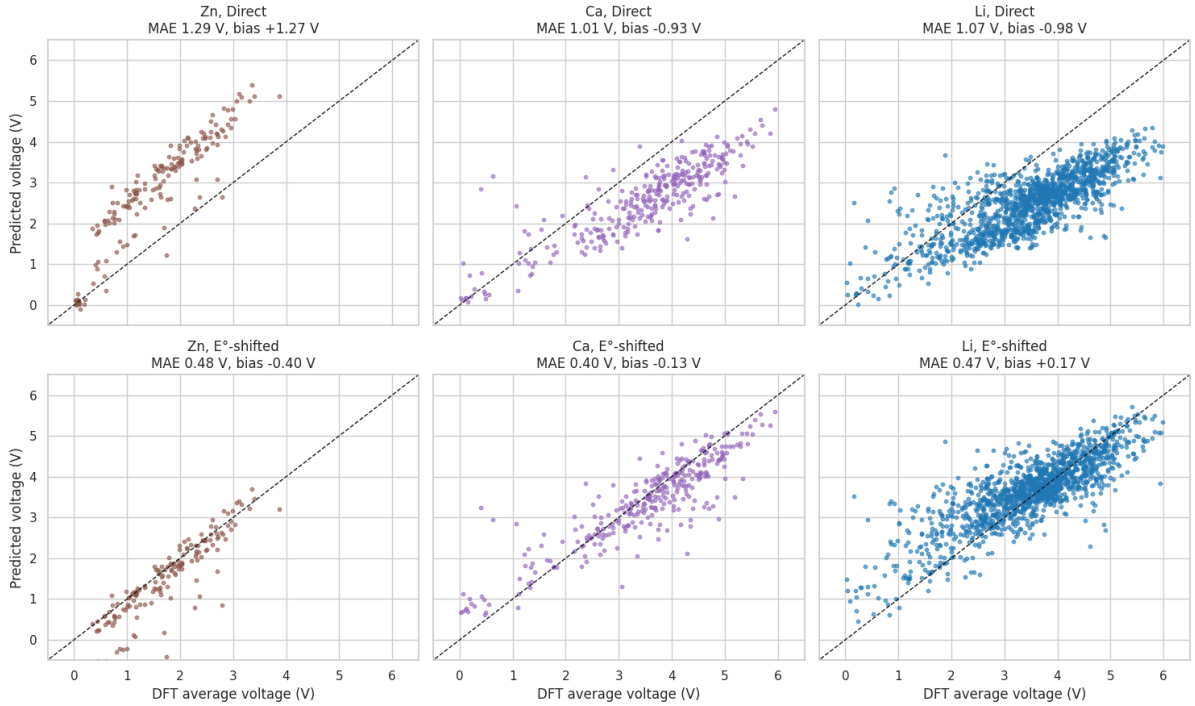


Figure 6.2: *Zero-shot parity for Zn, Ca and Li before (top) and after (bottom) the E° shift.*

6.4 No cost in-distribution

When all ions are in training, the E°-shifted model gives 0.368 V against 0.369 V for the direct model over the same 10 splits, so the correction can always be used. It is the model used in the application.

6.5 Significance

The shift is a simple, physically grounded change of target that lets a voltage model trained on common ions make useful first predictions for new ones (for example Rb, Cs or Al) before any DFT calculation. A literature check is needed before claiming novelty, since related reference-electrode corrections exist in electrochemistry.

The Electrode Voltage Predictor: An Interactive Tool

KEY RESULT

The final E°-shifted model is packaged as a Gradio web application that predicts the voltage, capacity and energy of a host from its chemical formula in under a second.

7.1 Using the application

The user types the charged host formula (e.g. `FePO4`, `MnO2`, `V2O5`, `Fe2(SO4)3`), selects a working ion and enters the number of ions inserted per formula unit. Worked examples are available with one click.

7.2 What the application returns

Table 7.1: *Outputs of the Electrode Voltage Predictor and how they are computed.*

Output	How it is computed
Predicted average voltage	Model predicts E_{host}; $V = E_{\text{host}} - E^\circ(\text{ion})$, set to 0 V if negative
Discharged formula	Host + x ions (e.g. $FePO_4$ + 1 Li $\rightarrow$ $LiFePO_4$)
Theoretical gravimetric capacity	$x \cdot z \cdot 26{,}801\ \text{mAh mol}^{-1}$ divided by the molar mass of the discharged formula
Theoretical specific energy	Voltage $\times$ capacity (Wh kg^{-1})
Redox-metal oxidation state	Charged $\rightarrow$ discharged, from the charge balance of Chapter 3
Trust message	$\pm$ MAE for trained ions; a zero-shot warning for Rb and Cs; an experimental warning for trivalent Al
Notes	Warnings for hosts without a redox metal, unreliable charge balance, unusually high oxidation states and clipped predictions
Materials Project reference	If the host and ion are in the database, the DFT voltage is shown for comparison

7.3 Robustness

The application validates its input. Invalid formulas (e.g. `XyZ2`, which crashed the first version), zero or negative ion counts, and hosts that already contain the working ion (e.g. `LiFePO4` for Li) return a helpful message instead of an error. Feature calculation is wrapped so that unusual formulas fail gracefully.

7.4 Files and deployment

The Week 5 notebook writes the application to Google Drive:

- `app.py` – the user interface;
- `featurize.py` – the same feature code used in training;
- `voltage_model_e0.json` – the trained XGBoost model;
- `app_metadata.json` – feature list and error figures;
- `reference_voltages.csv`, `requirements.txt` and a `README.md` with Hugging Face Spaces settings.

The application runs on Colab with a temporary public link and can be hosted permanently on Hugging Face Spaces and published on GitHub.

Chapter 8

Conclusions, Limitations and Future Work

8.1 Conclusions

- A composition-based XGBoost model predicts DFT electrode voltages of unseen host frameworks with an MAE of 0.445 V, half the error of a per-ion mean.
- Oxidation states from carefully validated charge balance improve this to 0.391 V; pretrained CHGNet structure fingerprints bring the best model to 0.368 ± 0.019 V ($R^2 = 0.82$).
- Na and K transfer between ions without loss, but Li, Ca and Zn fail by about 1 V, almost entirely through a constant offset tied to each ion's reference electrode.
- Training on $E_{\text{host}} = V + E^\circ$ removes 96–108% of that offset error without any target-ion data, halving the mean zero-shot error ($0.81 \rightarrow 0.44$ V) with no cost in-distribution.
- The model is available as an interactive web tool.

8.2 Limitations

- The targets are DFT (GGA/GGA+U) voltages, which carry their own systematic errors relative to experiment.
- Only average voltages are modelled; voltage profiles, kinetics and cycling stability are not.
- Anode-type hosts, very high voltages (above 5 V) and rare chemistries are under-represented.
- Y, Rb, Al and Cs had too little data to train or test properly, and trivalent ions are untested.
- The novelty of the E° shift still needs a literature check.

8.3 Future work

- Test zero-shot predictions for Rb, Cs and Al against their Materials Project entries.
- Add uncertainty estimates (quantile or ensemble models) to the application.
- Fine-tune a graph neural network directly on $E_{\text{host}} = V + E^\circ$.
- Validate selected predictions with new DFT calculations, for example in Quantum ESPRESSO.
- Release the code on GitHub and host the application on Hugging Face Spaces.

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